Indonesia International Institute for Life Sciences

[SESSION TWELVE: USE CASE]

Scalable SARS-CoV-2 Genome Analysis in R

A Shiny dashboard on a large genome collection

[BI2106] - [LINUX AND R PROGRAMMING]

[Arli Aditya Parikesit]
[arli.parikesit@i3l.ac.id]

Use Case: Scalable SARS-CoV-2 Surveillance

A lab holds a SARS-CoV-2 genome collection: a metadata table plus genome FASTA. It is too big for read.csv() and lm() in RAM. Four questions drive this session.

1 Lineage dynamics

How does lineage frequency change by month and country?

data.table, dplyr sampling

2 Genome features

Do length, N-content, GC and 3-mer profiles carry signal?

Biostrings, bigmemory

3 Sequencing quality

Does the N-fraction trend with time and platform?

chunked linear model (biglm idea)

4 Honest plotting

How do we draw every genome without a black blob?

hexbin, 2D density, sampling

Outcome: apply sampling, chunking, memory-mapped storage and streamed models, and drive them from a Shiny dashboard in RStudio.

Why This Is a Big-Data Problem

Metadata table

One row per genome: accession, collection date, location, lineage, platform. read.csv() parses every column into RAM, text columns are the expensive ones.

Genome FASTA

About 30,000 nucleotides per genome. With millions of records the sequence file, not the table, dominates disk and time. Never load it whole; stream it.

Rule of thumb from Session 12: share of RAM taken by the data

under 20%: workable	20-50%: large, chunk or sample	over 50%: massive, unusable in RAM

Three tactics from the session solve this: reduce what you read (select columns, sample), keep data on disk (ff, big.matrix), and process in chunks (streamed models).

Pipeline: Six Stages, One Script Each

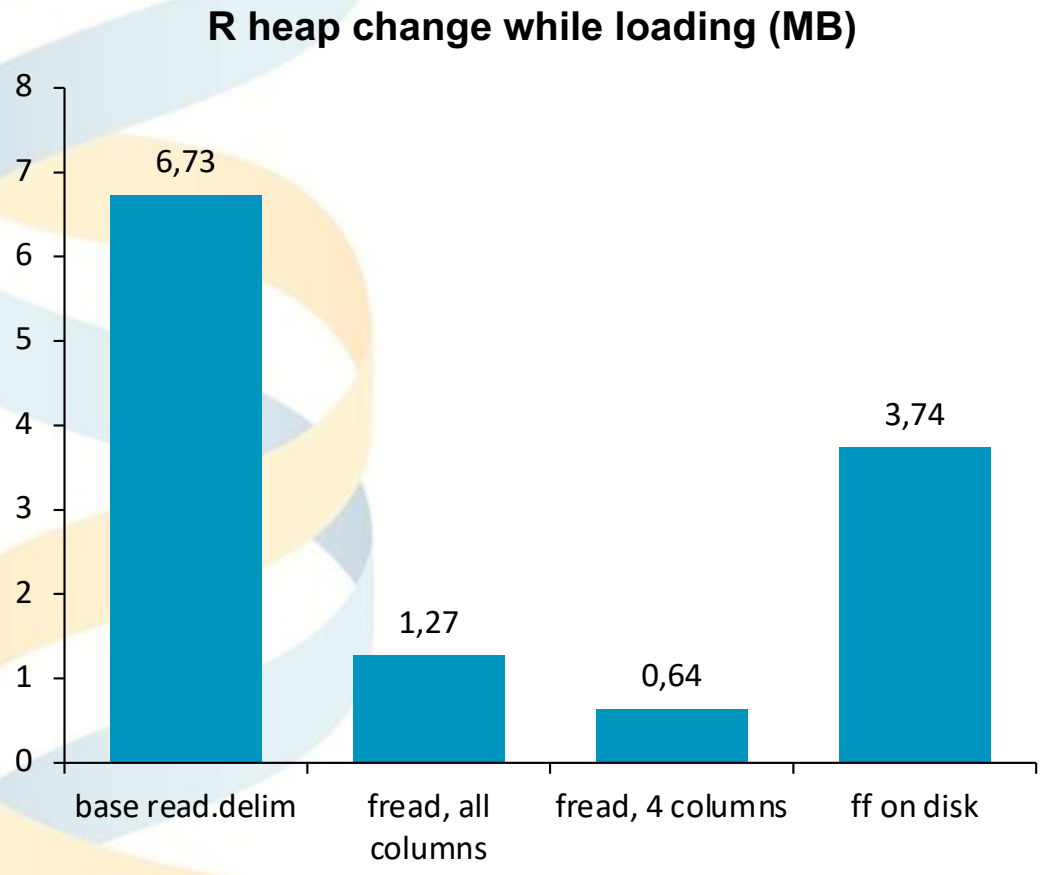
0 Acquire	1 Ingest	2 Sample	3 Features	4 Model	5 Dashboard
NCBI Virus download or synthetic generator `00_make_synthetic.R`	fread(select=), ff, memory baseline `01_ingest.R`	dplyr sample_frac, stratified `02_sample.R`	Biostrings chunks into big.matrix `03_features_bigmemory.R`	streamed linear model, plots `04_models_viz.R`	Shiny app in RStudio `app.R`

Session 12 link

Stage 1: slides 5, 10-12, 25-26. Stage 2: slides 6-8, 16. Stage 3: slides 19-25. Stage 4: biglm report. Plots: slides 13-17.

The dashboard calls the same helper functions as the scripts, so results match.

Stage 1: Load the Metadata Cheaply



Synthetic 20,000-row table, R 4.3.3, data.table 1.14.10, ff 4.0.12.

```
fread(f, select = keep)
read.delim(f)
ff::read.table.ffdf(
  file = f, sep = "\t")
```

What to read from this

Selecting 4 columns needs about half the heap of reading all of them with fread.

Base read.delim used the most.

Demo scale only: re-run on real data to see the gap grow.

Stage 2: How Wrong Is a 5% Sample?

Lineage	Full	Simple 5%	Stratified
Lineage_A	0.156	0.174	0.144
Lineage_B	0.197	0.201	0.203
Lineage_C	0.229	0.232	0.209
Lineage_D	0.203	0.192	0.205
Lineage_E	0.141	0.138	0.144
Lineage_F	0.075	0.063	0.095

```
sample_frac(as.data.frame(d), 0.05)
d[, .SD[sample(.N, ...)], by = .(month, country)]
```

Synthetic data. Simple sample n = 1,000; stratified n = 1,184 (at least one row per stratum).

Take-aways

Largest absolute error in a lineage share: 0.018 (simple) and 0.021 (stratified).

The smallest lineage (F, about 7%) is hit hardest.

Stratifying on month and country keeps time and place structure. It does not guarantee a better lineage estimate.

Rare lineages are where subsamples mislead.

Stage 3: Stream Genomes into a big.matrix

FASTA file
30.6 MB, 20,000 genomes

Chunk
2,000 genomes in RAM at a time

Features
length, N fraction, GC, 64 3-mers

big.matrix
20,000 x 67, 10.7 MB on disk

10.73 MB
R heap change to copy the matrix into RAM

0 MB
R heap change to attach it (memory-mapped)

```
idx <- fasta.index(fa)
x <- readDNAStringSet(idx[rows, ])
bm[rows, ] <- cbind(
  width(x), n_frac, gc,
  oligonucleotideFrequency(x, 3))
```

About 1.1 MB heap per 2,000-genome chunk, so memory stays flat as the file grows.

Stage 4: A Linear Model Without the Full Table

```
fit <- stream_fit(file, n_frac ~ t_years + platform, prep, chunk_rows = 5000)
```

Term	Estimate	SE	Generator direction
Intercept (Illumina, 2020)	0.00462	0.000097	baseline
Years since 2020-01	-0.00080	0.000042	negative
Ion Torrent vs Illumina	0.00319	0.000120	positive
Nanopore vs Illumina	0.00449	0.000090	positive

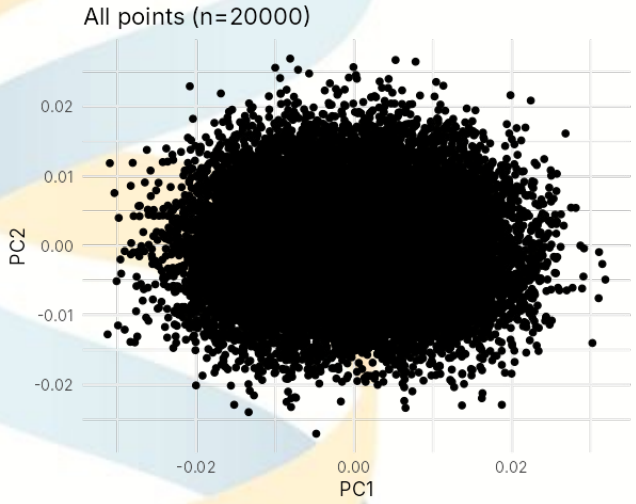
Validation

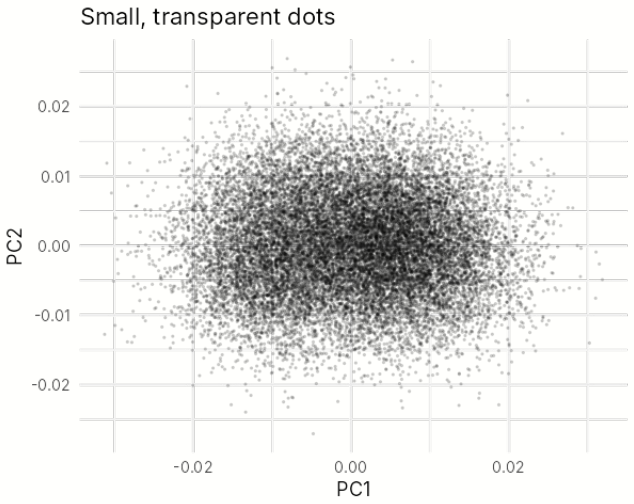
Largest coefficient gap, streamed vs lm(): 5.3e-16.

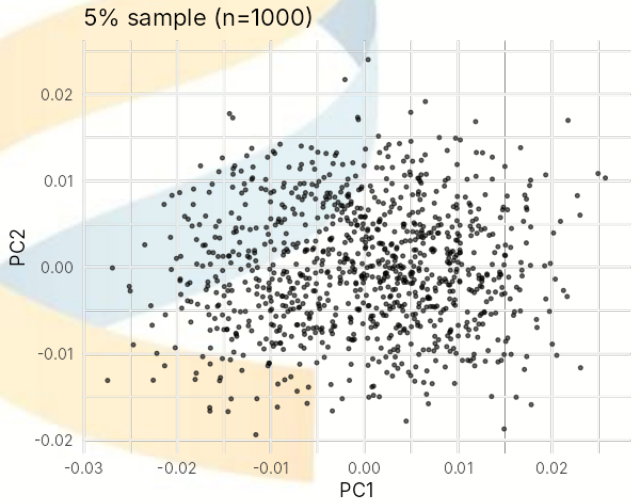
Engine and limit

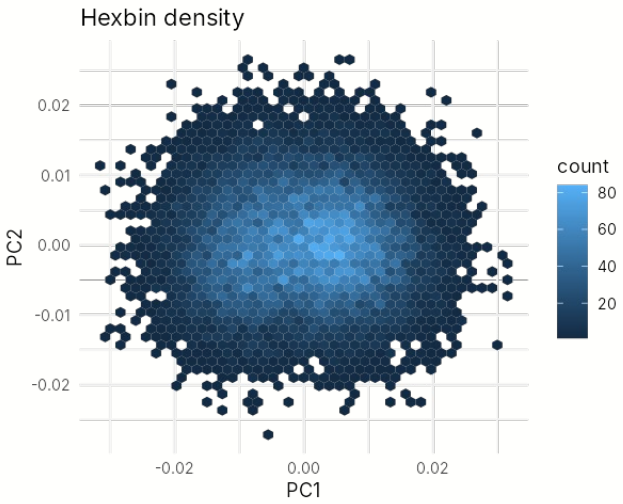
Normal-equation fallback used here. biglm path not run in the test sandbox.

Plot Every Genome, Avoid the Black Blob









Fixes from the session

Smaller, transparent dots.

Plot a 5% sample.

Bin the plane: hexbin or 2D density.

In the dashboard, switch the PCA plot mode and watch the draw time in the log.

PC1 and PC2 explain about 13% and 7% of variance. No lineage separation is claimed.

The Shiny Dashboard in RStudio



Lineage tab with 10% sampling (synthetic data).

Control map

Sample fraction
slides 6-8, 16

Lineage, country, dates
filters before sampling

Load backend
data.table, base R, ff

Chunk size
streamed model, 1,000 to 20,000 rows

PCA plot mode
all, sample, hexbin

Backends tab
RAM vs big.matrix, time log

Run scripts 00 to 04 once, then press Run App. Set SARS2_DATA_DIR to point the same dashboard at real data.

Pitfalls and Honest Limits

Synthetic data

Lineages, N fractions and k-mer shifts are simulated. Conclusions apply to the code and memory behaviour, not to SARS-CoV-2 biology.

Weak PCA signal

PC1 and PC2 explain about 13% and 7% of variance. No lineage separation is claimed.

Real column names

NCBI Virus headers may differ from the demo. Verify and edit CFG$cols before running real data.

Join held in RAM

The model table join keeps three small columns in memory. Very large tables need a database step.

Streaming precision

The normal-equation fallback loses precision with near-collinear predictors. biglm updates a QR decomposition.

Scale not measured

Demo is 20,000 genomes of 1.5 kb. Timings on full data and on your laptop were not measured here.

Group Extension Exercise

1. Download a real SARS-CoV-2 genome subset from NCBI Virus (for example Indonesian sequences), edit CFG$cols, and rerun scripts 01 to 04.

2. Set CFG$kmer = 4 and compare PCA variance explained with k = 3.

3. Install biglm and run stream_fit() with the biglm engine. Compare with the fallback.

4. Sweep the chunk size from 1,000 to 20,000 rows in the dashboard. Report seconds and heap change.

Deliver: dashboard screenshots, the version table, and a short note on what changed at real scale.

Summary and References

Constraints of large data are worked out by reading less, storing on disk, and processing in chunks.

A Shiny dashboard makes each choice visible with a time and memory log.

References

Mailund T (2017) Beginning Data Science in R. Apress.

Kane MJ, Emerson JW, Weston S (2013) Scalable strategies for computing with massive data. J Stat Softw 55(14):1-19. doi:10.18637/jss.v055.i14

de Jonge E, Wijffels J (2013) ffbase: statistical functions for large datasets. useR! 2013 abstract.

NCBI Datasets: Download a SARS-CoV-2 genome data package. https://www.ncbi.nlm.nih.gov/datasets/docs/how-tos/virus/get-sars2-genomes/

Tested with R 4.3.3, data.table 1.14.10, bigmemory 4.6.1, ff 4.0.12, Biostrings 2.70.2, shiny 1.8.0.